AI101 MINI-PROJECT

GLOBAL RENEWABLE ENERGY OUTPUT & WEATHER CORRELATION

Analysing the relationship between weather conditions and renewable energy generation.

REPO LINK:-

https://github.com/anishiitrpr/ai101_project82.git

Team Number: 82
Sehajpreet Singh Reen :- PPT Lead
Anish Kumar :- Video Lead
Dharmesh Meena :- Documentation
Parmeet :- Execution



PROBLEM & DATASET CONTEXT

WHAT ARE WE STUDYING?

Analysing the relationship between weather conditions and renewable energy generation, focusing on solar radiation, temperature, humidity, wind speed and cloud coverage.

WHY DOES IT MATTER?

With growing energy demands and environmental concerns, understanding these relationships can help improve energy planning and support the promotion of clean energy.

DATASET SOURCE

Renewable Power Generation and Weather Conditions via Kaggle.

Kaggle dataset link:

https://www.kaggle.com/datasets/pythonafroz/renewable-power-generation-and-weather-conditions/data

196,776

Total Records (15-min intervals, with some gaps)

17

Columns Recorded

14.91 MB

CSV File Size

2017–2022

1 Jan 2017 – 31 Aug 2022

PROJECT APPROACH & WORKFLOW

From raw Kaggle multi-variable records to structured correlation findings across 4 stages.



Data Input

Load the Kaggle CSV dataset and examine its 17 columns.



Data Inspection

Understand dataset structure, date coverage, missing intervals and data quality.



Linear Regression

Using the dataset from Kaggle.com
We simply trained our model using linear regression .
This model can now give output to new test data.



Data Visualization, Output & Evaluation

Present observed relationships through charts, graphs and meaningful visual comparisons. This helped us to train our model. Now this well-trained model can give output to new test-data.
For better performance we evaluated the accuracy of the model.

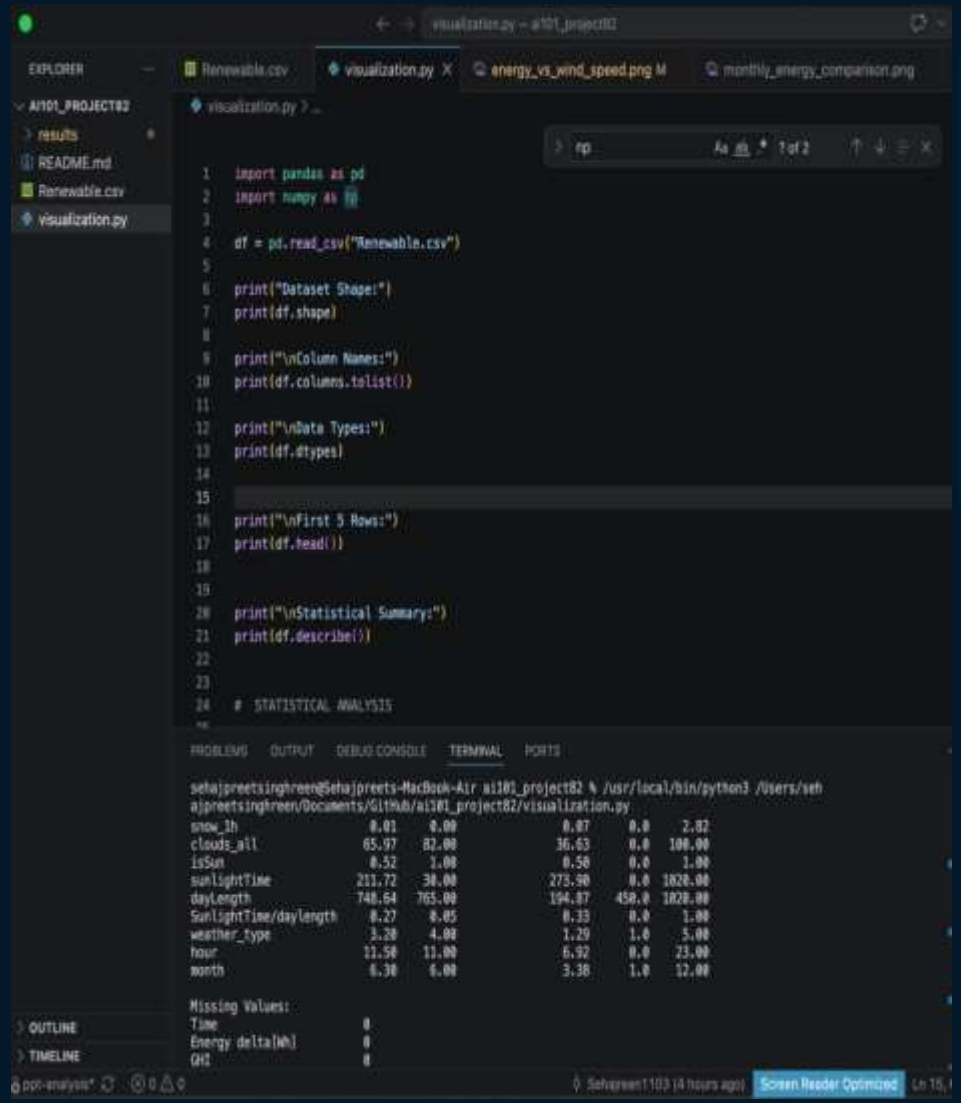
LIBRARIES USED:-

Pandas, NumPy, Matplotlib, Seaborn.

Core objective:-

To analyze how weather conditions influence global renewable energy output by identifying correlations, patterns, and trends in energy production.

RESULTS & DATASET INSIGHTS:



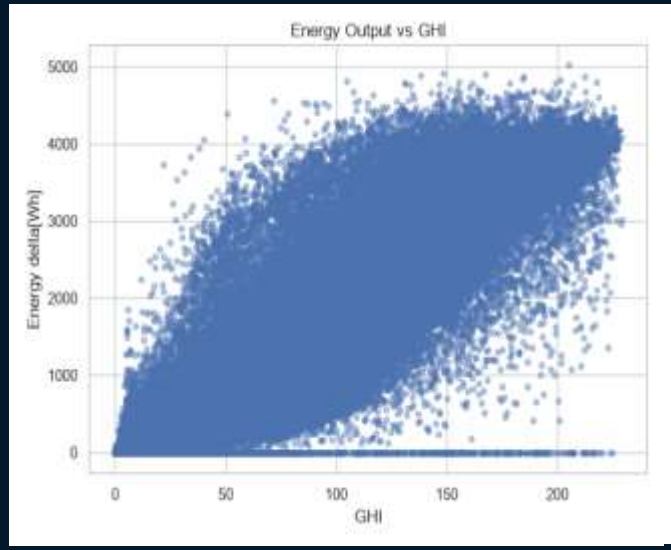
Key Numbers:-

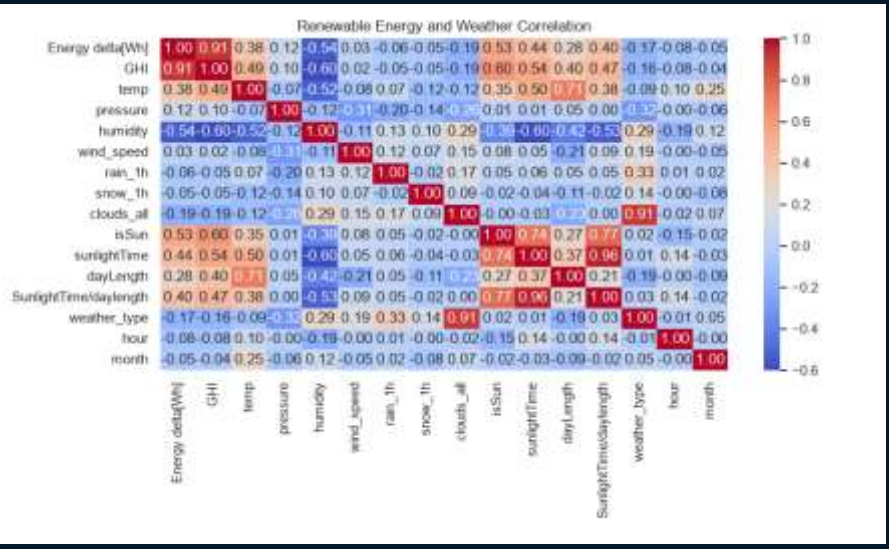
+0.91
GHI
Strong positive correlation

-0.54
HUMIDITY
Moderate negative correlation

+0.38
TEMPERATURE
Positive correlation

-0.19
CLOUD COVER
Weak negative correlation





KEY LEARNINGS & CONCLUSION

WHAT WORKED:

- Clear correlation patterns Between weather and energy output.
- Effective data visualization and comparisons.
- Identification of key influencing factors.

DATASET CHALLENGES (what didn't work):

- Inconsistent data.
- Weak correlations for some weather parameters.
- Limited ability to establish causation.

TEAM CONTRIBUTIONS & AI DISCLOSURE:

Member 1: Sehaj Reen:- PPT Lead

Member 2: Anish Kumar:- Video Lead

Member 3: Parmeet:- Execution

Member 4:- Dharmesh:- Documentation

AI Tool Usage:

ChatGPT, Claude & Gemini for references, understanding concepts, Python coding assistance, troubleshooting and GitHub guidance.

Conclusion: Understanding the relationship between weather conditions and renewable energy generation can contribute to better-informed clean energy planning.